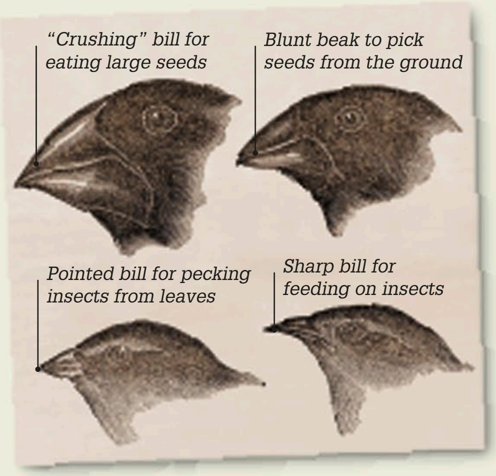
Understanding evolution

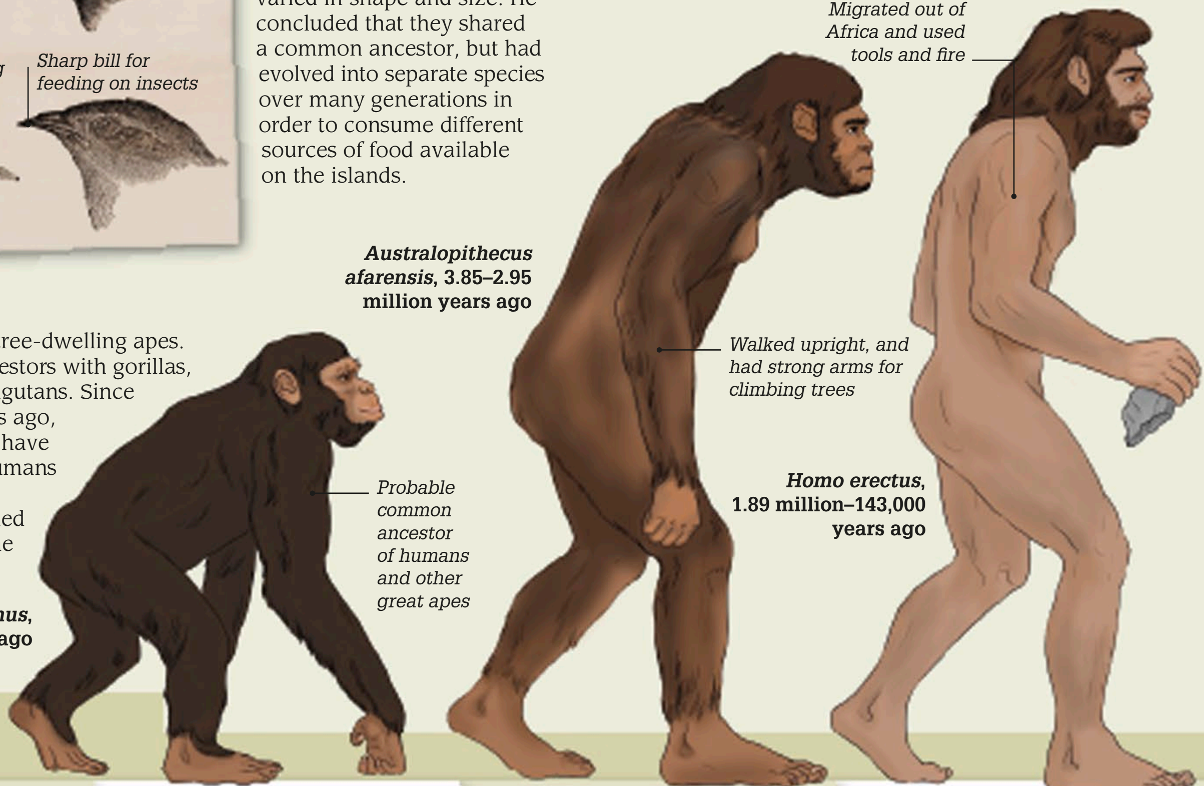
From the study of fossils, scientists in the 1800s concluded that life on Earth had changed slowly, or evolved, over billions of years from a single, simple ancestor into the millions of species that exist today. English naturalist Charles Darwin (see pp. 134–135) suggested that this had come about through natural selection. The modern study of genetics has proved him right by explaining the biological mechanisms that drive evolution.



Darwin's finches
Charles Darwin found different species of finch living on different islands of the Galápagos, (islands off the coast of Ecuador in South America). The finches' bills varied in shape and size. He concluded that they shared a common ancestor, but had evolved into separate species over many generations in order to consume different sources of food available on the islands.

Sexual selection
Rather than passing on a trait that helps an animal survive, sometimes what drives selection is being more attractive to a mate. A peacock, for example, fans out his dazzling tail when he is courting. It seems that peahens choose mates that have a bigger, brighter plumage. As a result, the peacock with the showiest feathers passes on his genes to the greatest number of offspring. This leads, over time, to more colorful birds.

Human evolution
Humans evolved from tree-dwelling apes. We share common ancestors with gorillas, chimpanzees, and orangutans. Since about four million years ago, all our direct ancestors have walked on two legs. Humans are the only existing member of a group called *Homo*; our species name is *Homo sapiens*.



Key events

1809

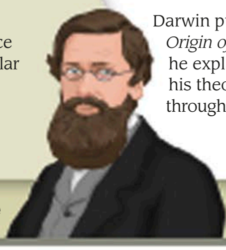
French naturalist Jean-Baptiste Lamarck argued that living creatures could pass on acquired characteristics to their offspring.

1830

Scottish geologist Charles Lyell showed that Earth had gone through many geological ages over millions of years, paving the way for Darwin's theory of evolution.

1858

English naturalist and explorer Alfred Russel Wallace independently developed similar ideas to Charles Darwin's on natural selection.


Alfred Russel Wallace

1859

Darwin published *On the Origin of Species*, in which he explained in detail his theory of evolution through natural selection.